Clinical Decision Support Systems (Telehealth, Telemedicine and Teledentistry)

Reference: Chapter 16 Robert E. Hoyt. Medical informatics-Practical guide for health care professionals, 3rd Edition, 2009

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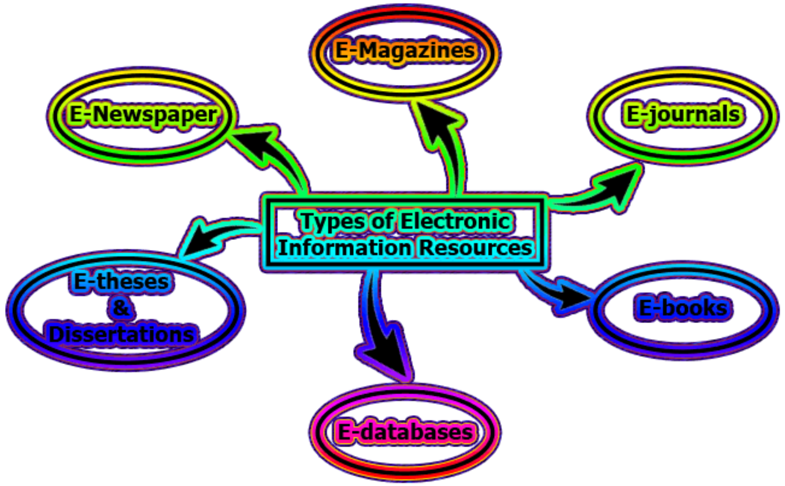
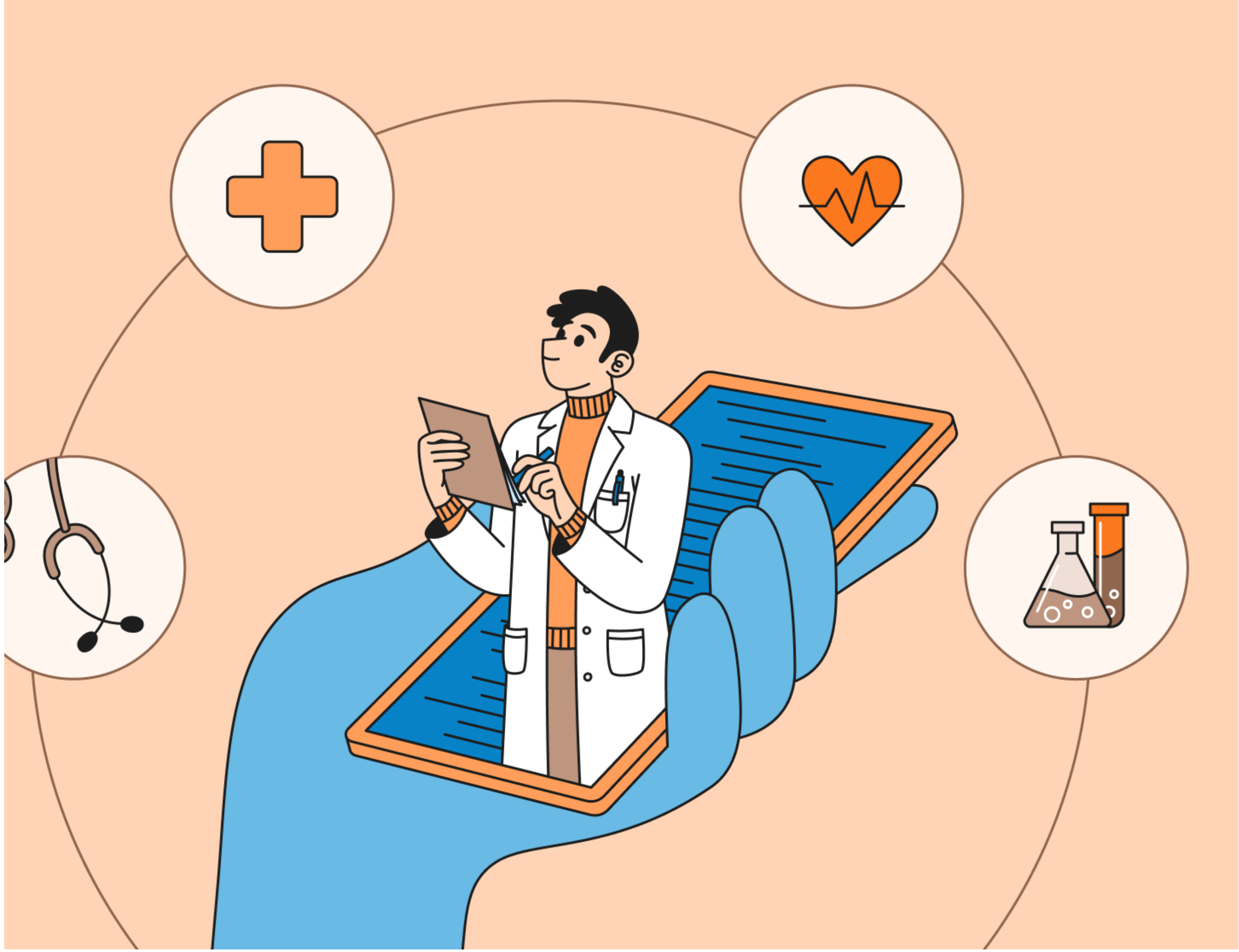


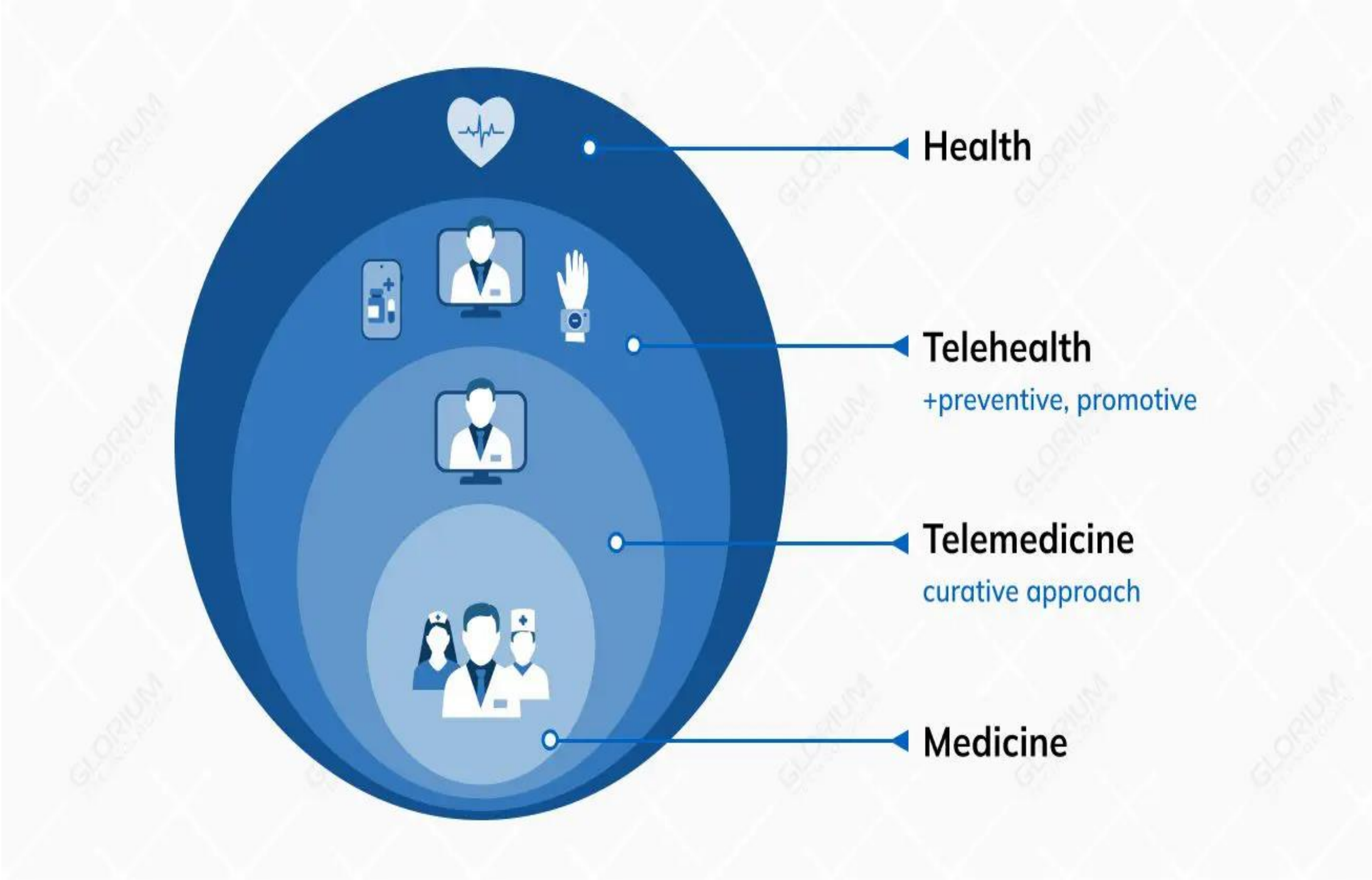
Definition: Telehealth

"The use of **electronic information** and **telecommunications technologies** to support **long-distance** clinical health care, **patient** and **professional health-related education**, **public health** and **health administration**"



Definition: Telemedicine

"The use of **medical information** exchanged from **one site** to another via **electronic communications** to **improve patients' health** status"



Goal of Telemedicine

1. To provide **timely and high quality medical care remotely**
2. **High patient satisfaction** due to better access to specialty care and less time lost from work.

Early Telemedicine (Courtesy Radio News)

Telemedicine Communication Modes

Communication Mode	Pros	Cons
Patient-Portal secure-Messaging	Asynchronous. Able to attach photos. Response can be formatted with template. Could use VoIP. Audit trail	Not as personal as live visit. Usually not connected to EHR or other information
Telephone	Widely available, simple and inexpensive	Not asynchronous. Unstructured. No audit trail
Audio-Video	Maximal input to clinician. Can include review of x-rays, etc. Perhaps more personal than just messaging	Currently, most expensive in terms of networks and hardware but that is changing

Telemedicine Transmission Modes

1. Store-and-forward
2. Real time
3. Remote monitoring

1. Store-and-forward

Images or videos are saved and sent later. As an example, a primary care physician takes a picture of a rash with a digital camera and forwards it to a dermatologist to view when time permits.

This could also be referred to as asynchronous communication.

2. Real time

A specialist at a medical center views video images transmitted from a remote site and discusses the case with a physician. This requires more sophisticated equipment to send images real time and often involves two-way interactive televisions.

This would be an example of synchronous communication.

3. Remote monitoring

A technique to monitor patients at home, in a nursing home or in a hospital for personal health information or disease management.

Telemedicine Categories

- Teleradiology,
- Teledermatology,
- Telecardiology,
- Telepathology,
- Telesurgery,
- Telepsychiatry,
- Teleneurology,
- Teleophthalmology,
- Telepharmacy

1. Traditional Telemedicine

- Robot Rounds
- E-ICU Rounding

2. Telerounding of inpatients

- Health Buddy
- HoneyWell HomMed
- Telemanagement

3. Telehomecare and Telemanagement

1. Traditional Telemedicine

Most programs consist of a central medical hub and several rural spokes. Programs attempt to improve access to services in rural and underserved areas, to include prisons.

Teleradiology, Teledermatology, Telecardiology, Telepathology, Telesurgery, Telepsychiatry, Teleneurology, Teleophthalmology, Telepharmacy, etc

2. Telerounding

This is a new concept developed to help address the shortage of physicians and nurses.

Telerounding is an advanced telemedicine service that enables healthcare providers to conduct patient rounds remotely, improving the efficiency and effectiveness of hospital and clinic operations.

2. Telerounding

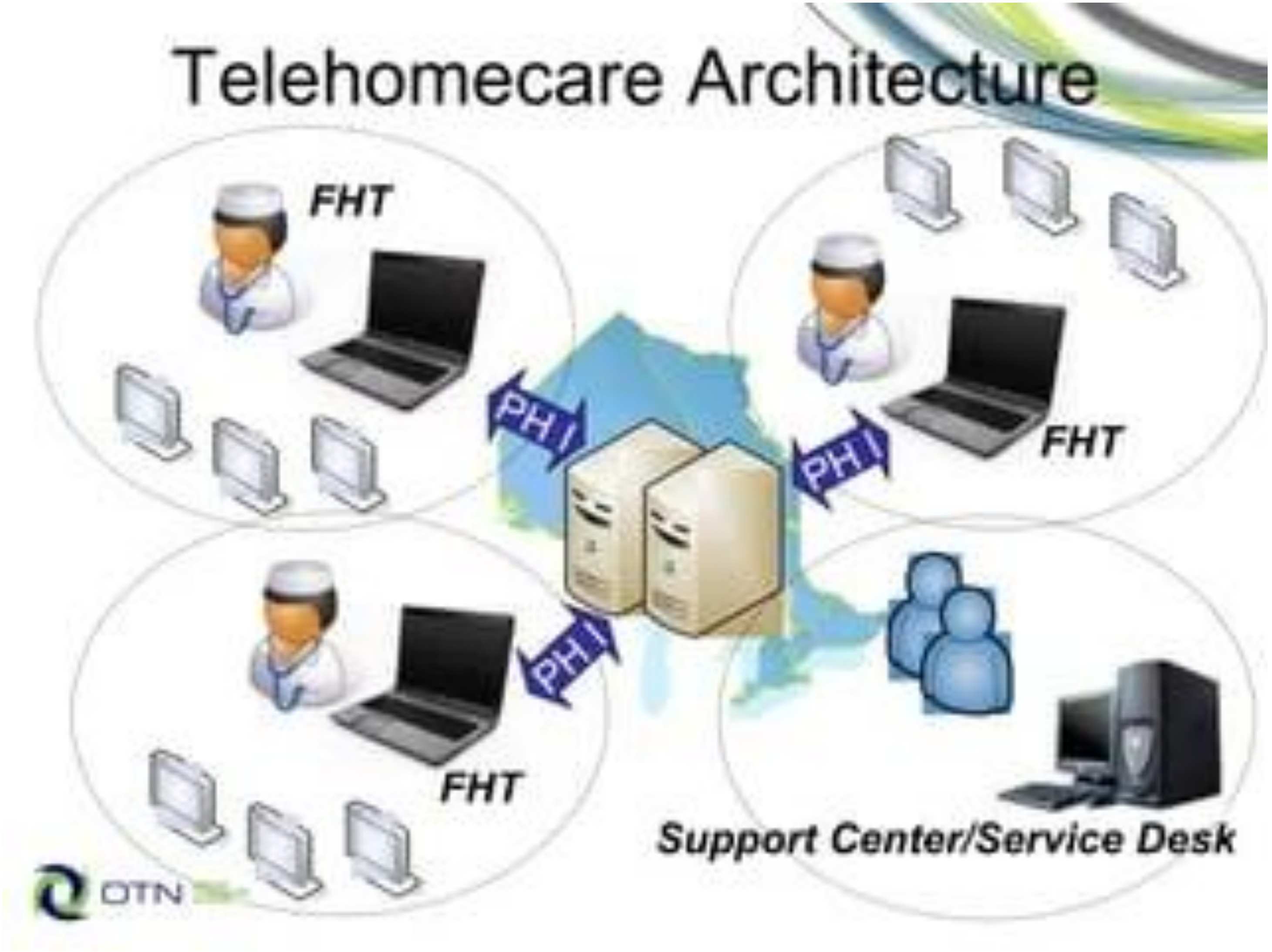
Key Features of Telerounding Software

1. **Scheduling and Alerts:** Automates rounding schedules and sends alerts to ensure no check-ins are missed.
2. **Real-Time Data Access:** Allows immediate access to patient charts, lab results, and diagnostic data, facilitating informed decision-making from remote locations.

3. Telehomecare and Telemanagement

Telemonitoring programs allow for audio and visual contact with patients at home and therefore they can save a visit by a nurse or physician.

Instead of nurses making routine home visits, then can make visits only if there is a problem, such as a change in symptoms or vital signs.



Telehomecare Program

1. Health Buddy
2. HoneyWell HomMed

Health Buddy

- Example of a popular home monitoring system
- Data is sent via phone lines
- Device comes with desktop decision support software
- It covers 45 disease protocols
- Program is interactive with patients; it asks questions daily

HoneyWell HomMed

- Provides voice messages to patients in multiple languages
- Standard and optional features: digital weight scale, blood pressure, oximetry, glucometer,
- Data is transmitted via phone lines
- LifeStream Connect™ is a new option that interfaces with EHRs

Telemanagement

Programs will be **interactive** and include **patient education** for issues such as drug compliance.

This data may **interface** with an **electronic health record**, **information system** or **web site** for others to evaluate.

Some predict that **houses** will be **wired** with multiple small sensors known as “**motes**” that will monitor daily activities such as taking medications and leaving the house.

Telehealth Organizations and Research

OFFICE FOR THE ADVANCEMENT OF TELEHEALTH (OAT)

AMERICAN TELEMEDICINE ASSOCIATION (ATA)

Barriers to Telemedicine

- Limited reimbursement
- Slow clinical acceptance because of the newness of the technology
- Limited research showing reasonable benefit and return on investment
- High cost or the limited availability of high-speed telecommunications
- Bandwidth issues, particularly in rural areas where telemedicine is most needed.
- Lack of standards
- Fear of malpractice as a result of telemedicine

Teledentistry

Similar to telemedicine, has emerged as a new tool with promising benefits for various dental disciplines including endodontics, orthodontics, oral surgery and pediatric dentistry.

It holds the potential to improve access to, and delivery of oral healthcare in rural and underserved areas.

Almazrooa SA, Mansour GA, Alhamed SA, Ali SA, Akeel SK, Alhindi NA, Felemban OM, Mawardi HH, Binmadi NO. The application of teledentistry for Saudi patients' care: A national survey study. J Dent Sci. 2021 Jan;16(1):280-286.

Definition: Teledentistry

Teledentistry is a combination of telecommunications and dentistry involving the exchange of clinical information and images over remote distances for dental consultation and treatment planning.

Jampani ND, Nutalapati R, Dontula BS, Boyapati R. Applications of teledentistry: A literature review and update. J Int Soc Prev Community Dent. 2011 Jul;1(2):37-44.

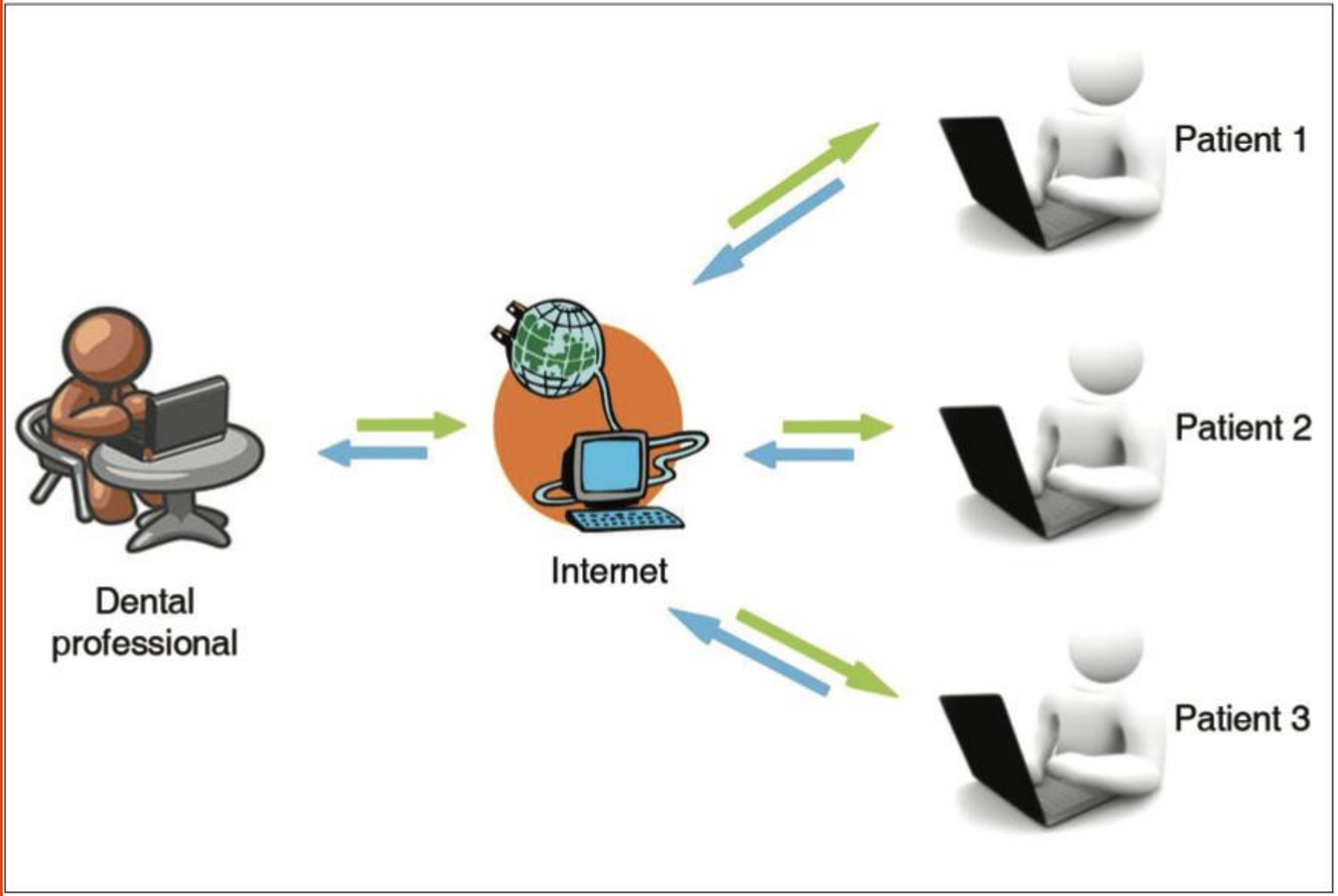
Basis of Teledentistry

Internet is the basis of modern systems of Teledentistry.

Jampani ND, Nutalapati R, Dontula BS, Boyapati R. Applications of teledentistry: A literature review and update. J Int Soc Prev Community Dent. 2011 Jul;1(2):37-44.

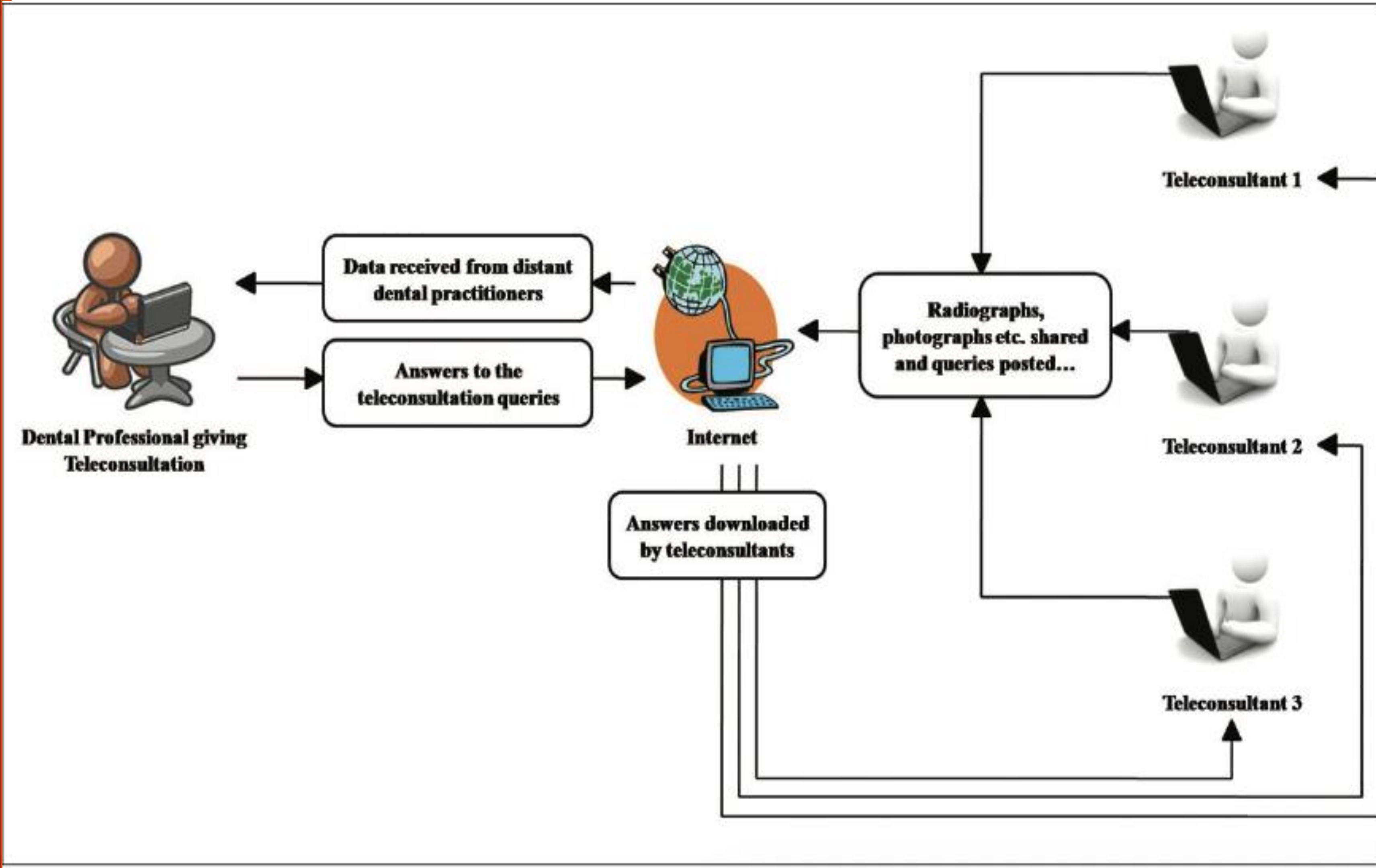
Methods of teleconsultation in dentistry: Real-time consultation

Jampani ND, Nutalapati R, Dontula BS, Boyapati R. Applications of teledentistry: A literature review and update. J Int Soc Prev Community Dent. 2011 Jul;1(2):37-44.



Methods of teleconsultation in dentistry: Store-and-forward method

Jampani ND, Nutalapati R, Dontula BS, Boyapati R. Applications of teledentistry: A literature review and update. J Int Soc Prev Community Dent. 2011 Jul;1(2):37-44.



Ethical and legal issues

Concerns about the **confidentiality of dental information** arise from the transfer of medical histories and records

Practitioners of teledentistry should take utmost care to ensure that **patient privacy** is not compromised by unauthorized entities.

Patients should be made aware that their information is to be transmitted electronically, and the **possibility exists that the information will be intercepted**, despite maximum efforts to maintain security.

Jampani ND, Nutalapati R, Dontula BS, Boyapati R. Applications of teledentistry: A literature review and update. J Int Soc Prev Community Dent. 2011 Jul;1(2):37-44.

Conclusion

Telemedicine is still in its infancy in most areas of the country.

The barriers are largely financial due to the high cost to set up the system and the lack of reimbursement in many cases.

Fortunately, the price of telemedicine systems is dropping so it may eventually be cheaper to use telemedicine in rural areas than to refer patients to distant urban specialists.

Thank you